

CHE 201 Fall 2019 HW 9

Giselle Cuevas

TOTAL POINTS

98 / 100

QUESTION 1

11 15 / 15

✓ - **0 pts** Correct

- **15 pts** no submission
- **6 pts** v2 calculation incorrect
- **6 pts** P2 calculation incorrect

QUESTION 2

22 15 / 15

✓ - **0 pts** Correct

- **15 pts** Missing answer
- **1 pts** Wrong calculation. $W = -25.8 \text{ kW}$
- **3 pts** $h_1 = 417.46 \text{ kJ/kg}$. $W = -25.8 \text{ kW}$
- **6 pts** $(z_1 - z_2) = -25 \text{ m}$ (negative). $W = -25.8 \text{ kW}$
- **3 pts** $h_3 = 3230.9 \text{ kJ/kg}$. Mass flow rate of steam = 0.0616 kg/s

QUESTION 3

33 20 / 20

✓ - **0 pts** Correct

- **5 pts** power requirement = $-W$
- **20 pts** No submission
- **8 pts** wrong calculation for W
- + **5 pts** Extra points from Question 6

QUESTION 4

44 13 / 15

- **0 pts** Correct

- **15 pts** Missing answer

✓ - **2 pts** Did not define the control volumes

- **2 pts** Did not state the engineering model
- **3 pts** Should have used m_2 to calculate work. $W = 199.9 \text{ hp}$
- **3 pts** $h_1 \sim h_f(180 \text{ }^\circ\text{C}) = 763.22 \text{ kJ/kg}$. $m_2 = 0.371 \text{ kg/s}$

- **1 pts** $m_2 = ?$

- **1 pts** Work is positive

QUESTION 5

55 15 / 15

✓ - **0 pts** Correct

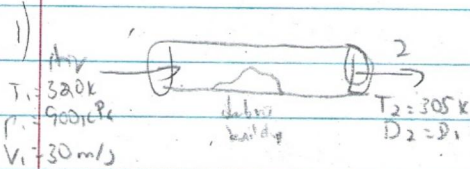
- **15 pts** Missing or wrong answer
- **2 pts** $W_{1-2} = 0.002 \text{ kJ}$
- **7.5 pts** b) $t = 1328.3 \text{ s}$ (22.1 min)
- **2 pts** $W_{2-3} = -0.0522 \text{ kJ}$
- **2 pts** $h_e = 2675.5 \text{ kJ/kg}$
- **3 pts** b) calculation is not clear.
- **13 pts** Incomplete answer
- **2 pts** $u_2 = 417.36 \text{ kJ/kg}$
- + **6 pts** Extra points from Question 6

QUESTION 6

66 20 / 20

✓ - **0 pts** Correct

- **20 pts** Missing answer
- + **20 pts** Extra points for making the board
- **10 pts** Missing or incomplete explanation of how it works and/or missing schematics of important parts (eg: nozzle, fan, switch, heating elements, cord etc).
- **5 pts** Missing source for the drawing
- **2 pts** Missing schematics of the air flow
- **10 pts** Missing drawing



model

one inlet - 1 outlet

steady state $\rightarrow$ ideal gasconstant diameter ($D_1 = D_2$)

$$\dot{Q}_{cv} = 0$$

$$PE = 0$$

$$W_w = 0$$

Energy Balance

$$\cancel{\frac{dZ_{cv}}{dt}} \rightarrow \cancel{\dot{Q}_{cv}} = \cancel{\dot{W}_w} + \dot{m} \left[(h_1 - h_2) + \frac{(V_1^2 - V_2^2)}{2} + g(z_1 - z_2) \right]$$

$$0 = \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} \right]$$

a) @ $T = 320 \text{ K} \rightarrow h_1 = 320.29 \text{ kJ/kg}$

@ $T = 305 \text{ K} \rightarrow h_2 = 305.22 \text{ kJ/kg}$

$$0 = \left(320.29 \frac{\text{kJ}}{\text{kg}} - 305.22 \frac{\text{kJ}}{\text{kg}} \right) + \left(\frac{30^2 \frac{\text{m}^2}{\text{s}^2} - V_2^2}{2} \right)$$

$$\frac{-30^2 + V_2^2}{2} = 15.07 \frac{\text{kJ}}{\text{kg}}$$

$$-30^2 + V_2^2 = 2(15.07)$$

$$V_2^2 = (30^2) + 2(15.07 \frac{\text{kJ}}{\text{kg}})$$

$$V_2 = \sqrt{(30)^2 + 2(15.07 \frac{\text{kJ}}{\text{kg}} \cdot 1000 \frac{\text{J}}{\text{kJ}})}$$

$$\boxed{V_2 = 176.182 \text{ m/s}}$$

b) $\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$

$$p_2 = \frac{p_1 V_1 T_2}{T_1 V_2}$$

$$p_2 = \frac{(900 \text{ kPa})(30 \text{ m/s})(305 \text{ K})}{(320 \text{ K})(176.182 \text{ m/s})}$$

$$\boxed{p_2 = 146.067 \text{ kPa}}$$

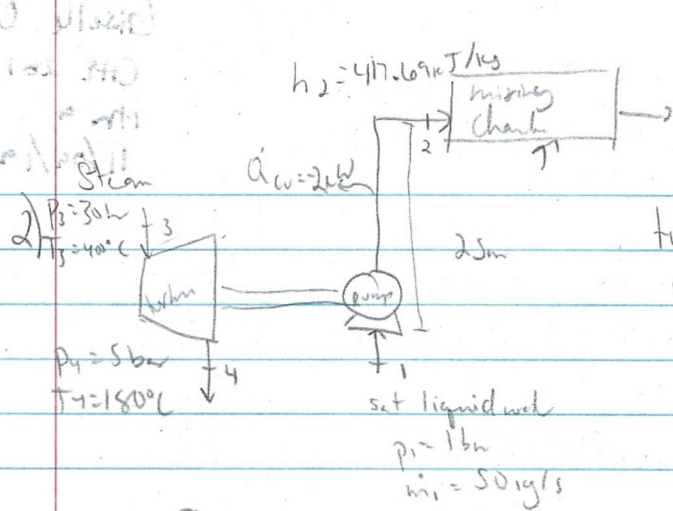
11 15 / 15

✓ - 0 pts Correct

- 15 pts no submission

- 6 pts v2 calculation incorrect

- 6 pts P2 calculation incorrect



$$a) \frac{dE_{cv}}{dt} = \dot{Q}_w - \dot{W}_w + \dot{m}_1 \left[h_1 + \frac{V_1^2}{2} + z_1 g \right] - \dot{m}_2 \left[h_2 + \frac{V_2^2}{2} + z_2 g \right]$$

$$\dot{Q}_w + \dot{m}_1 [h_1 + z_1 g] = \dot{W}_w + \dot{m}_2 [h_2 + z_2 g]$$

$$-2 \text{ kW} + 50 \frac{\text{kg}}{\text{s}} \left[417.69 \frac{\text{kJ}}{\text{kg}} + (0 \times 9.81) \right] = \dot{W}_w \left[417.69 \frac{\text{kJ}}{\text{kg}} + (25 \text{ m} \times 9.81 \frac{\text{m}}{\text{s}^2} \cdot \frac{1}{1000}) \right]$$

$$\dot{W}_w = -25.76 \text{ kW}$$

$$b) \text{ @ } 30 \text{ bar \& } 400^\circ\text{C} \rightarrow h_3 = 3230.9 \text{ kJ/kg}$$

$$\text{ @ } 5 \text{ bar \& } 180^\circ\text{C} \rightarrow h_4 = 2812 \text{ kJ/kg}$$

$$\frac{dE_{cv}}{dt} = \dot{Q}_w - \dot{W}_w + \dot{m} \left[(h_3 - h_4) + \frac{V_3^2 - V_4^2}{2} + g(z_3 - z_4) \right]$$

$$\dot{W}_w = \dot{m} (h_3 - h_4)$$

$$\frac{25.76 \text{ kW}}{3230.9 - 2812 \frac{\text{kJ}}{\text{kg}}} = \dot{m}$$

$$\dot{m} = 0.0615 \text{ kg/s}$$

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✓ - 0 pts Correct

- 15 pts Missing answer

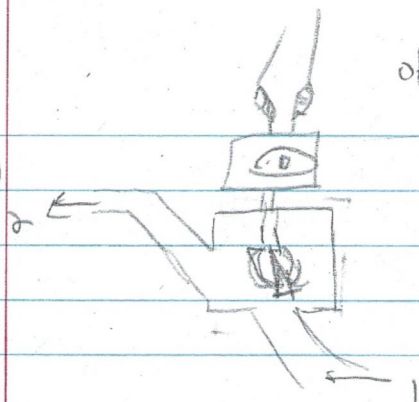
- 1 pts Wrong calculation. $W = -25.8 \text{ kW}$

- 3 pts $h_1 = 417.46 \text{ kJ/kg}$. $W = -25.8 \text{ kW}$

- 6 pts $(z_1 - z_2) = -25\text{m}$ (negative). $W = -25.8 \text{ kW}$

- 3 pts $h_3 = 3230.9 \text{ kJ/kg}$. Mass flow rate of steam = 0.0616 kg/s

3)

off pump
electric motor

$$Q = 150 \text{ m}^3/\text{s}$$

$$\Delta KE = 0$$

$$\Delta z = 20$$

$$Q_w = -0.6 \text{ m}^3/\text{s}$$

$$\Delta p = 0$$

$$\Delta T = 0$$

$$\dot{m} = \rho Q$$

$$\dot{m} = 1000 \text{ kg/m}^3 \cdot 150 \text{ m}^3/\text{s}$$

$$\dot{m} = 150,000 \text{ kg/s}$$

$$\frac{dE}{dt} = Q - W + m \left[(h_1 - h_2) + \frac{(V_1^2 - V_2^2)}{2} + g(z_1 - z_2) \right]$$

$$W = Q + m [g(\Delta z)]$$

$$W = 0.6 \text{ kW} \left(\frac{10^6 \text{ W}}{\text{kW}} \right) + 150,000 \left(9.81 \times 20 \right)$$

$$W = 3.003 \times 10^7 \text{ W} \times \frac{1 \text{ MW}}{10^6 \text{ W}}$$

$$\boxed{W = 30.03 \text{ MW}}$$

The power will be less than the pump power because the power is proportional to the volumetric flow rate at any member. Loss of power from friction.

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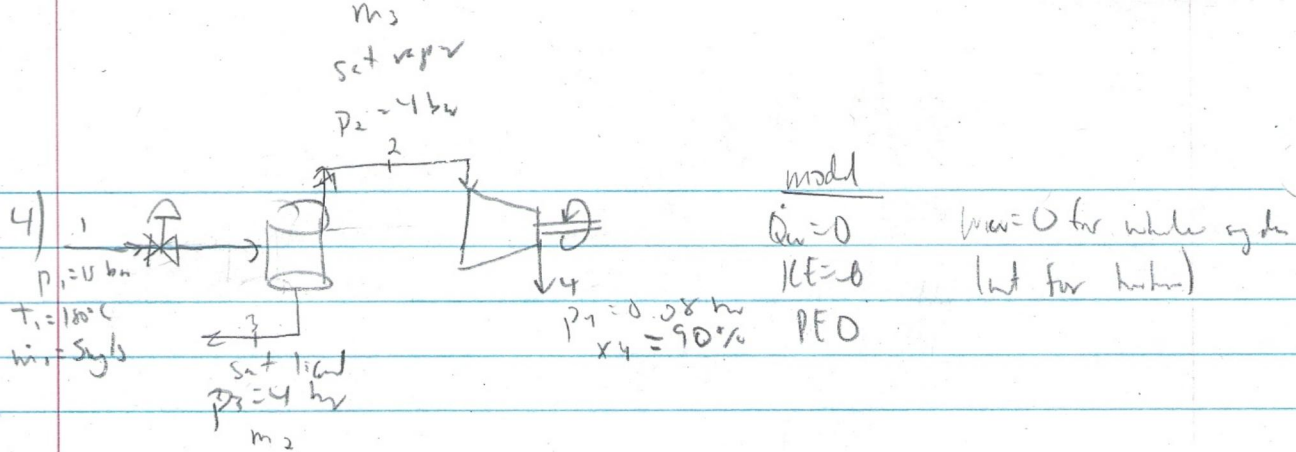
✓ - 0 pts Correct

- 5 pts power requirement = -W

- 20 pts No submission

- 8 pts wrong calculation for W

+ 5 pts Extra points from Question 6



Mass Balance

$$m_1 = m_2 + m_3$$

$$m_2 = m_1 - m_3$$

$$m_3 = m_1 - m_2$$

$$\frac{dE_{cv}}{dt} = \dot{Q}_w - \dot{W}_w + m_1 \left[h_1 + \frac{V_1^2}{2} + g z_1 \right] - m_2 \left[h_2 + \frac{V_2^2}{2} + g z_2 \right] - m_3 \left[h_3 + \frac{V_3^2}{2} + g z_3 \right]$$

$$0 = \dot{m}_1 h_1 - \dot{m}_2 h_2 - \dot{m}_3 h_3$$

$$\dot{m}_1 h_1 = \dot{m}_2 h_2 + \dot{m}_3 h_3$$

$$\dot{m}_1 h_1 = \dot{m}_2 h_2 + (\dot{m}_1 - \dot{m}_2) h_3$$

$$\dot{m}_1 h_1 = \dot{m}_2 h_2 + \dot{m}_1 h_3 - \dot{m}_2 h_3$$

$$\dot{m}_1 h_1 - \dot{m}_1 h_3 = \dot{m}_2 h_2 - \dot{m}_2 h_3$$

$$\dot{m}_2 = \frac{\dot{m}_1 (h_1 - h_3)}{(h_2 - h_3)}$$

$$@ T_1 = 180^\circ\text{C} \rightarrow h_f \approx h_1 = 763.22 \text{ kJ/kg}$$

$$@ p_2 = 4 \text{ bar} \rightarrow h_g \approx h_2 = 2738.6 \text{ kJ/kg}$$

$$@ p_4 = 0.08 \text{ bar} \rightarrow h = h_f + x(h_g - h_f)$$

$$h = 173.88 + 0.9(2577 - 173.88)$$

$$h_4 = 2336.69 \text{ kJ/kg}$$

$$@ p_3 = 4 \text{ bar} \rightarrow h_f \approx h_3 = 604.74 \text{ kJ/kg}$$

$$m_2 = \frac{5 \left(763.22 - 604.74 \right)}{(2738.6 - 604.74)}$$

$$m_2 = 0.371 \text{ kg/s}$$

$$\frac{dE_{cv}}{dt} = \dot{Q}_w - \dot{W}_w + m_2 \left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right]$$

$$\dot{W}_w = m_2 (h_2 - h_1)$$

$$\dot{W}_w = 0.371 \left(2738.6 - 2336.69 \frac{\text{kJ}}{\text{kg}} \right) \times 1 \text{ hp}$$

$$0.7457 \text{ kW}$$

$$\dot{W}_w = 199.96 \text{ hp}$$

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- 0 pts Correct
- 15 pts Missing answer
- ✓ - 2 pts Did not define the control volumes
 - 2 pts Did not state the engineering model
 - 3 pts Should have used m_2 to calculate work. $W = 199.9$ hp
 - 3 pts $h_1 \sim h_f(180 \text{ oC}) = 763.22$ kJ/kg. $m_2 = 0.371$ kg/s
 - 1 pts $m_2 = ?$
 - 1 pts Work is positive

5) mass balance $\frac{dm_{cv}}{dt} = \dot{m}_1 - \dot{m}_2 \rightarrow 0$
 $\dot{m}_1 = \dot{m}_2$

$V = 0.5L$

$T_1 = 20^\circ C$

$p_1 = 1 \text{ bar}$

$\dot{Q} = 0.85 \text{ kW}$

a) $m = \frac{V}{v} = 0.5L$

@ $T_1 = 20^\circ C \rightarrow v_1 \approx v_f = 1.0018 \times 10^{-3} \text{ m}^3/\text{kg}$ and $u_1 \approx u_f = 83.96 \text{ kJ/kg}$
 $m = \frac{0.5}{1.0018 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}} = 0.501 \text{ kg}$

@ $p_1 = 1 \text{ bar}$ $u_2 \approx u_g = 417.36 \text{ kJ/kg}$ and $v_2 \approx v_g = 1.0432 \times 10^{-3}$
 $W = m \int p dv$

$W = m \times p (v_2 - v_1)$
 $W = 0.5 \text{ kg} \cdot 1 \text{ bar} \cdot \frac{100 \text{ kPa}}{1 \text{ bar}} \cdot \frac{1000 \text{ N/m}^2}{1 \text{ kPa}} \cdot (1.0432 \times 10^{-3} - 1.0018 \times 10^{-3} \text{ m}^3/\text{kg}) \cdot \frac{1 \text{ kJ}}{1000 \text{ Nm}}$

$W = 0.00207 \text{ kJ}$

$\dot{Q} = \dot{U} + \dot{W}$

$\dot{Q} \Delta t = m(u_2 - u_1) + W$

$\Delta t = \frac{m(u_2 - u_1) + W}{\dot{Q}}$

$\Delta t = \frac{0.5L (417.36 \frac{\text{kJ}}{\text{kg}} - 83.96 \frac{\text{kJ}}{\text{kg}}) + 0.00207 \text{ kJ}}{0.85 \text{ kW}}$

$\Delta t = 196.18 \text{ sec}$

b) mass balance

$\frac{dm_{cv}}{dt} = -\dot{m}_e$

$\frac{dU_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} - \dot{m}_e h_e$

$\frac{dU_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \frac{dm_{cv}}{dt} h_e$

$\Delta U_{cv} = \dot{Q}_{cv} \Delta t - \dot{W}_{cv} \Delta t + \Delta m_{cv} h_e$

$m_3 u_3 - m_2 u_2 = \dot{Q}_{cv} \Delta t - \dot{W}_{cv} \Delta t + (m_3 - m_2) h_e$

$$m_2 = m_1 = 0.5 \text{ kg}$$

$$m_3 = 0.1 \text{ kg}$$

$$-m_2 u_2 = Q_w \Delta t - W_w - m_2 h_1$$

$$\Delta t = \frac{m_2 (h_1 - u_2) + W_w}{Q_w}$$

$$@ 1 \text{ bar } h_1 \approx h_g = 2675.5 \text{ kJ/kg}$$

$$W_w = \int p dV = p(V_1 - V_2) = -pV_2 = -m_2 p_2 v_2$$

$$W_w = -(0.5 \text{ kg})(1 \text{ bar}) \left(\frac{100 \text{ kPa}}{1 \text{ bar}} \right) \left(\frac{1000 \frac{\text{N}}{\text{m}^2}}{1 \text{ kPa}} \right) \left(1.0432 \times 10^{-3} \frac{\text{m}^3}{\text{kg}} \right) \left(\frac{1 \text{ kJ}}{1000 \frac{\text{N} \cdot \text{m}}{\text{m}}}} \right)$$

$$W_w = -0.052 \text{ kJ}$$

$$\Delta t = \frac{0.5 \text{ kg} \cdot (2675.5 \text{ kJ/kg} - 417.36 \frac{\text{kJ}}{\text{kg}}) + (-0.052 \text{ kJ})}{2.8 \text{ kW}} \cdot \left(\frac{1 \text{ kW}}{1 \frac{\text{kJ}}{\text{s}}} \right)$$

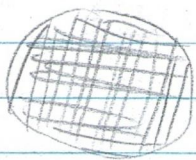
$$\Delta t = 1328.26 \text{ sec}$$

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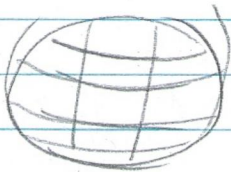
✓ - 0 pts Correct

- 15 pts Missing or wrong answer
- 2 pts $W_{1-2} = 0.002 \text{ kJ}$
- 7.5 pts b) $t = 1328.3 \text{ s}$ (22.1 min)
- 2 pts $W_{2-3} = -0.0522 \text{ kJ}$
- 2 pts $h_e = 2675.5 \text{ kJ/kg}$
- 3 pts b) calculation is not clear.
- 13 pts Incomplete answer
- 2 pts $u_2 = 417.36 \text{ kJ/kg}$
- + 6 pts Extra points from Question 6

6)



Wire net - keeps thing from getting inside the dryer



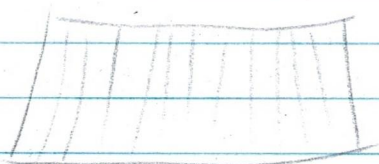
Vent - air goes in



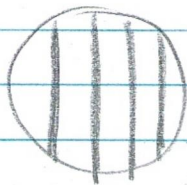
Switch - controls how much air goes through (low, med, high)



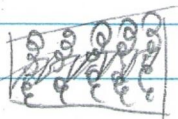
fan - cools down heated coils



Mica sheet : protects the outside plastic from the heat while the dryer is on



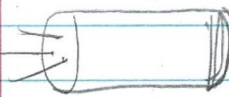
front wire net - prevents things from getting inside the dryer



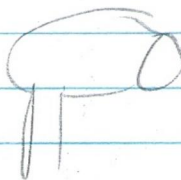
heater with coils - heats up when electricity passes through. Also acts as a resistor to control voltage that goes to the motor

 bimetallic strip - made up of 2 different metals, shut
down electricity to heater when it gets too hot

 thermal fuse - fan will blow and shut down the dryer
when it gets too hot

 motor - turn on the fan inside of
the dryer

 electric wire - runs the electricity and
powers the system

 → outside plastic case -
hold everything inside

 nozzle - concentrates air flow

6 6 20 / 20

✓ - 0 pts Correct

- 20 pts Missing answer

+ 20 pts Extra points for making the board

- 10 pts Missing or incomplete explanation of how it works and/or missing schematics of important parts (eg: nozzle, fan, switch, heating elements, cord etc).

- 5 pts Missing source for the drawing

- 2 pts Missing schematics of the air flow

- 10 pts Missing drawing